

MEDICAL INSURANCE DATA ANALYSIS PROJECT

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Setting a Working directory

```
setwd("C:/Users/BOBBY/Desktop/R Programming")
```

Data Importation

```
gss <- read.csv("insurance_data.csv")  
head(gss)
```

```
##   age    sex    bmi children smoker   region   charges  
## 1  19 female 27.900         0    yes southwest 16884.924  
## 2  18   male 33.770         1     no southeast 1725.552  
## 3  28   male 33.000         3     no southeast 4449.462  
## 4  33   male 22.705         0     no northwest 21984.471  
## 5  32   male 28.880         0     no northwest 3866.855  
## 6  31 female 25.740         0     no southeast 3756.622
```

```
tail(gss)
```

```
##      age    sex    bmi children smoker   region   charges  
## 1333  52 female 44.70         3     no southwest 11411.685  
## 1334  50   male 30.97         3     no northwest 10600.548  
## 1335  18 female 31.92         0     no northeast 2205.981  
## 1336  18 female 36.85         0     no southeast 1629.833  
## 1337  21 female 25.80         0     no southwest 2007.945  
## 1338  61 female 29.07         0    yes northwest 29141.360
```

```
str(gss)
```

```
## 'data.frame':   1338 obs. of  7 variables:  
##  $ age      : int  19 18 28 33 32 31 46 37 37 60 ...  
##  $ sex      : chr   "female" "male" "male" "male" ...  
##  $ bmi      : num   27.9 33.8 33 22.7 28.9 ...  
##  $ children: int    0 1 3 0 0 0 1 3 2 0 ...  
##  $ smoker   : chr    "yes" "no" "no" "no" ...  
##  $ region   : chr   "southwest" "southeast" "southeast" "northwest" ...  
##  $ charges  : num  16885 1726 4449 21984 3867 ...
```

Data Cleaning

Check for missing values

```
missing_value <- sum(is.na(gss))  
print(missing_value)
```

```
## [1] 0
```

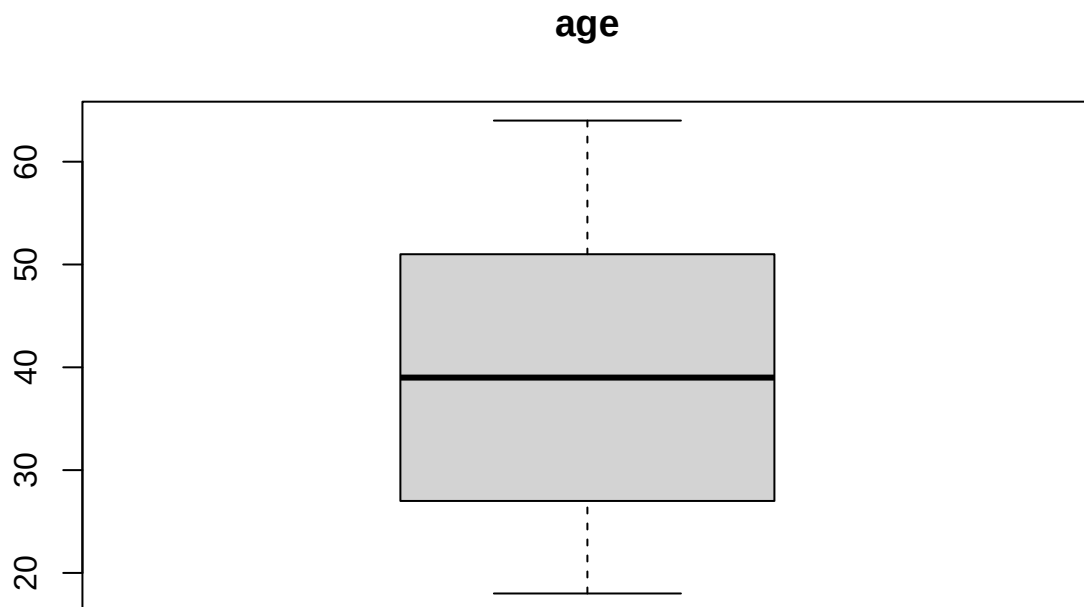
```
sum(duplicated(gss))
```

```
## [1] 1
```

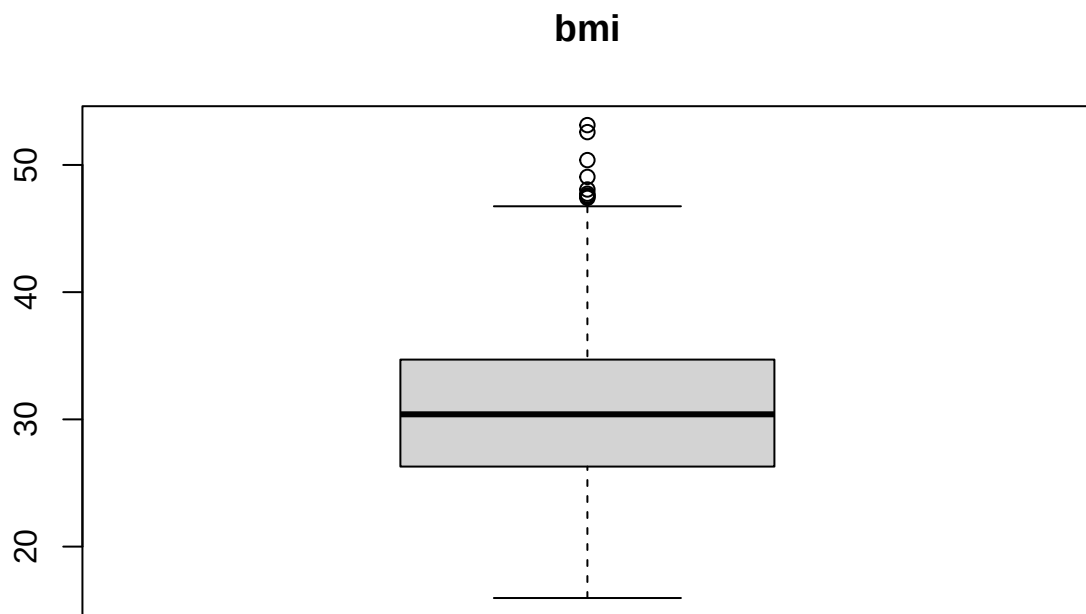
```
gss <- gss[!duplicated(gss),]
```

Detecting Outliers

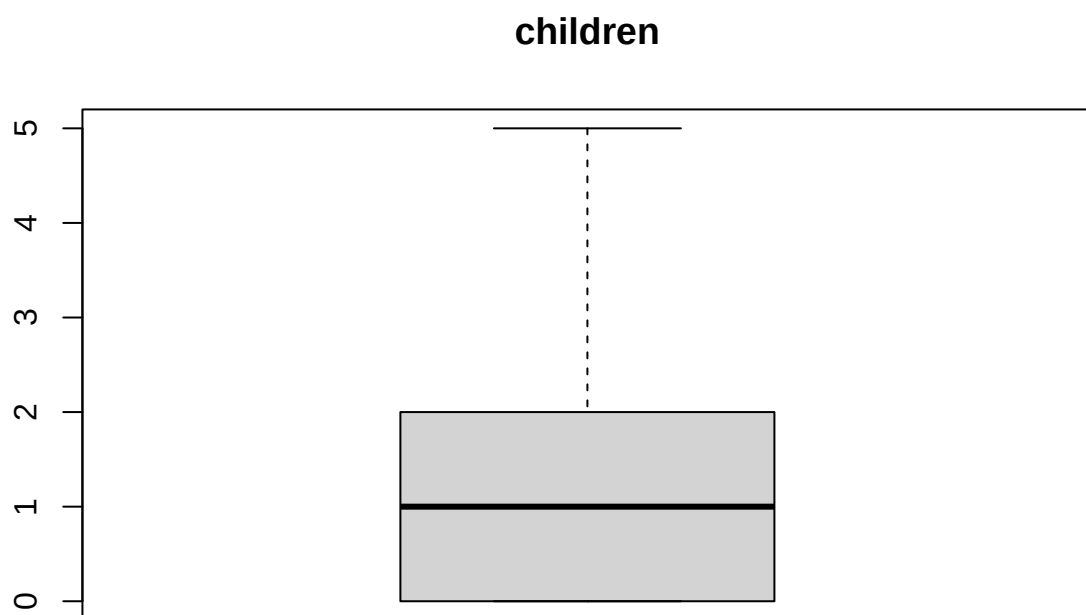
```
boxplot(gss$age, main = "age")
```



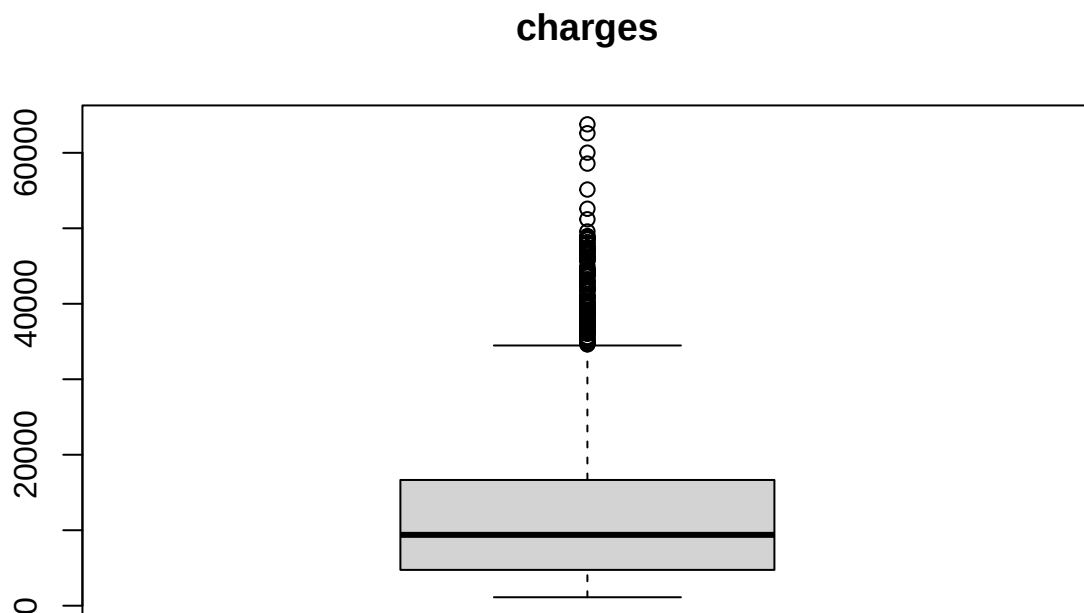
```
boxplot(gss$bmi, main = "bmi")
```



```
boxplot(gss$children, main = "children")
```



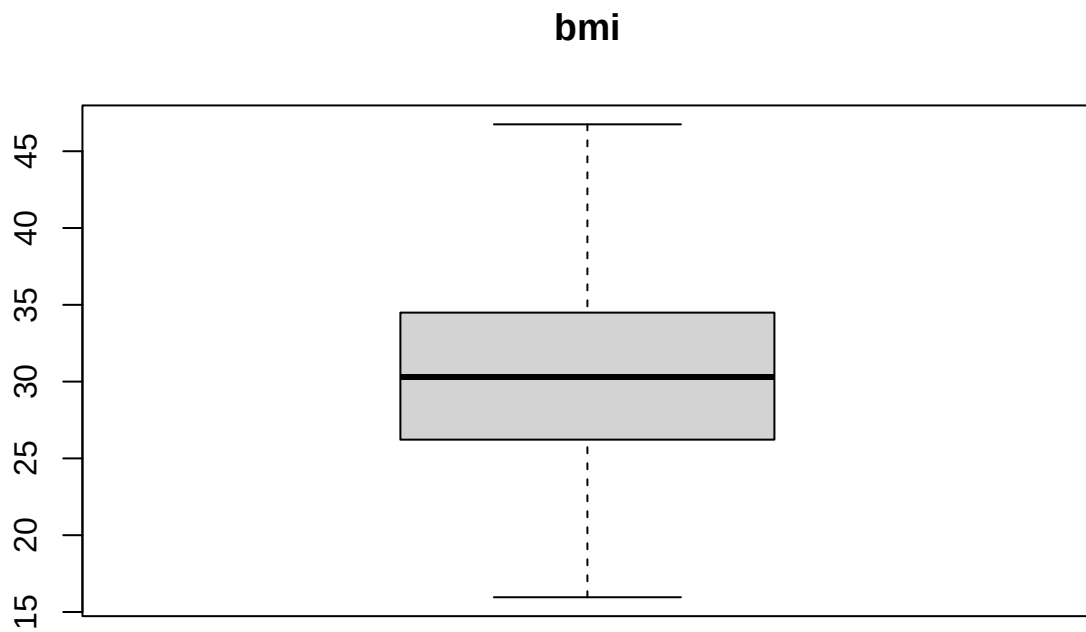
```
boxplot(gss$charges,main = "charges")
```



Handling outliers using 1.5 IQR Rule

```
Q1 <- quantile(gss$bmi, 0.25)
Q3 <- quantile(gss$bmi, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR
d_cleaned <- gss[gss$bmi >= lower_bound & gss$bmi <= upper_bound, ]
```

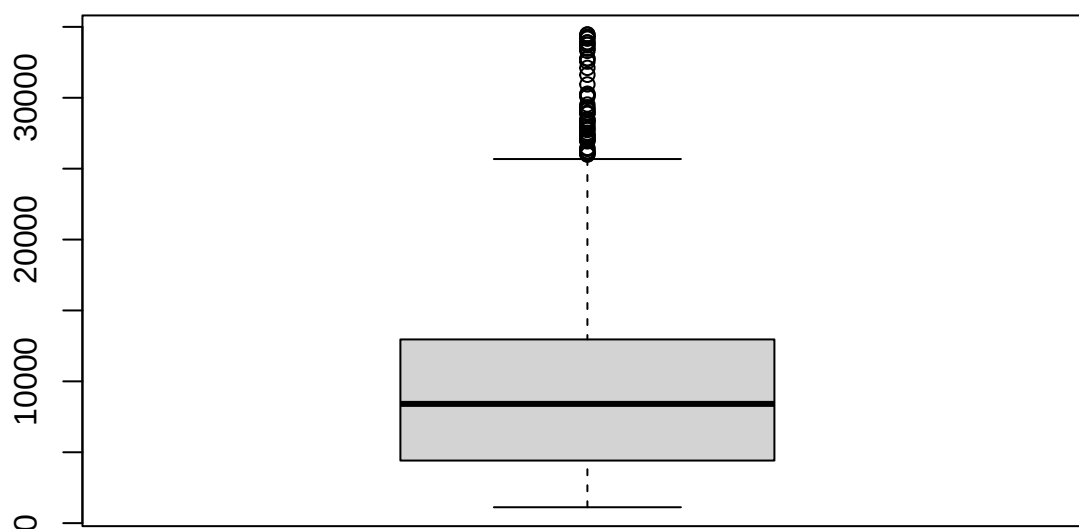
```
boxplot(d_cleaned$bmi, main = "bmi")
```



```
Q1 <- quantile(gss$charges, 0.25)
Q3 <- quantile(gss$charges, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR
d_cleaned <- gss[gss$charges >= lower_bound & gss$charges <= upper_bound, ]
```

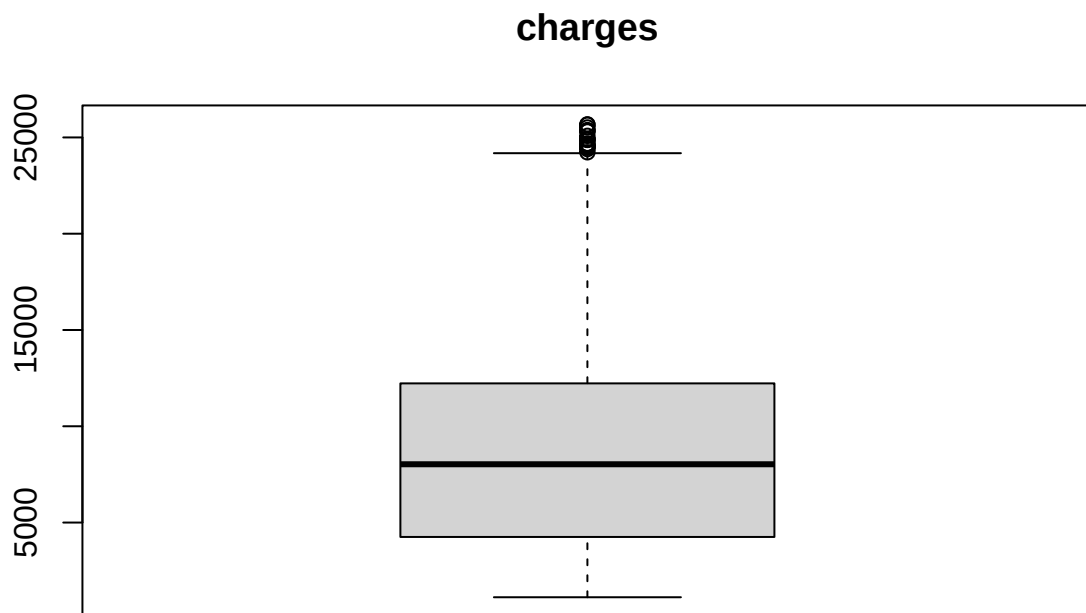
```
boxplot(d_cleaned$charges, main = "charges")
```

charges



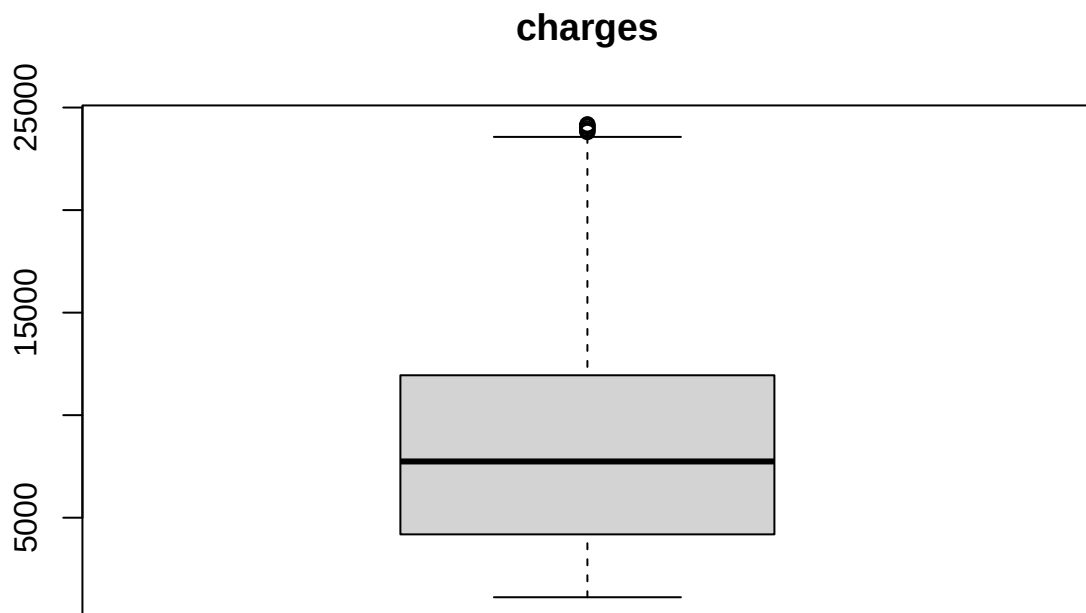
```
Q1 <- quantile(d_cleaned$charges, 0.25)
Q3 <- quantile(d_cleaned$charges, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR
g_cleaned <- d_cleaned[d_cleaned$charges >= lower_bound & d_cleaned$charges <= upper_bound, ]

boxplot(g_cleaned$charges, main = "charges")
```



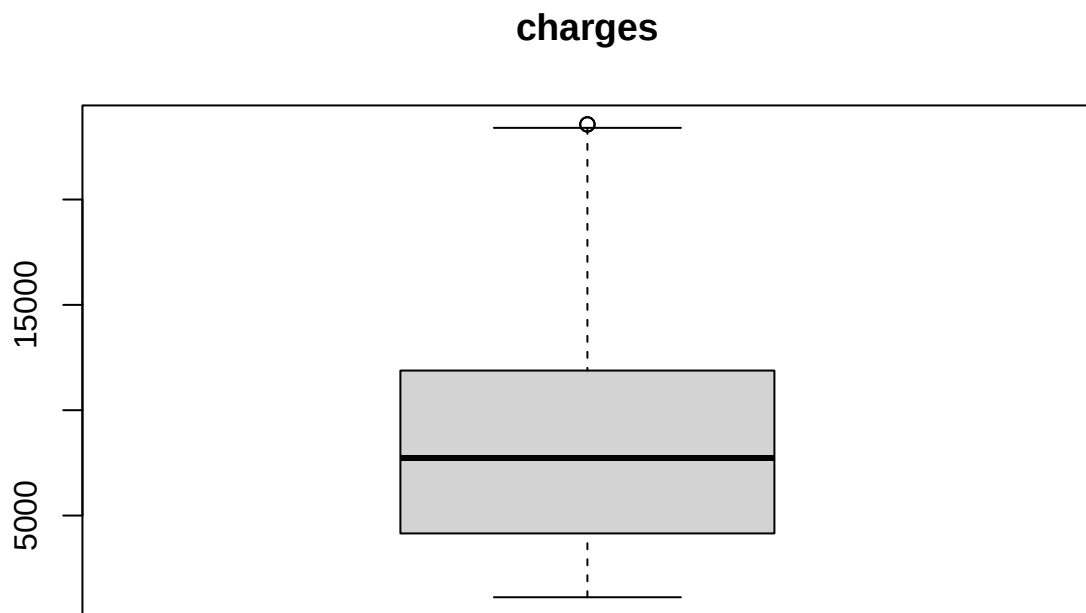
```
Q1 <- quantile(g_cleaned$charges, 0.25)
Q3 <- quantile(g_cleaned$charges, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR
s_cleaned <- g_cleaned[g_cleaned$charges >= lower_bound & g_cleaned$charges <= upper_bound, ]

boxplot(s_cleaned$charges, main = "charges")
```

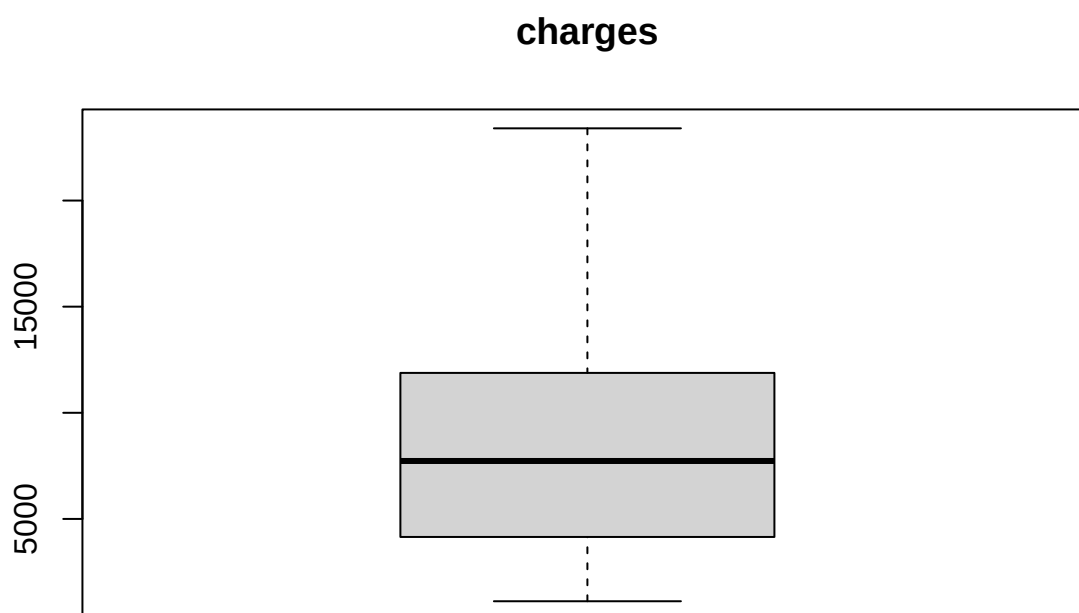
```
Q1 <- quantile(s_cleaned$charges, 0.25)
Q3 <- quantile(s_cleaned$charges, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR
t_cleaned <- s_cleaned[s_cleaned$charges >= lower_bound & s_cleaned$charges <= upper_bound, ]

boxplot(t_cleaned$charges, main = "charges")
```



```
Q1 <- quantile(t_cleaned$charges, 0.25)
Q3 <- quantile(t_cleaned$charges, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR
r_cleaned <- t_cleaned[t_cleaned$charges >= lower_bound & t_cleaned$charges <= upper_bound, ]

boxplot(r_cleaned$charges, main = "charges")
```



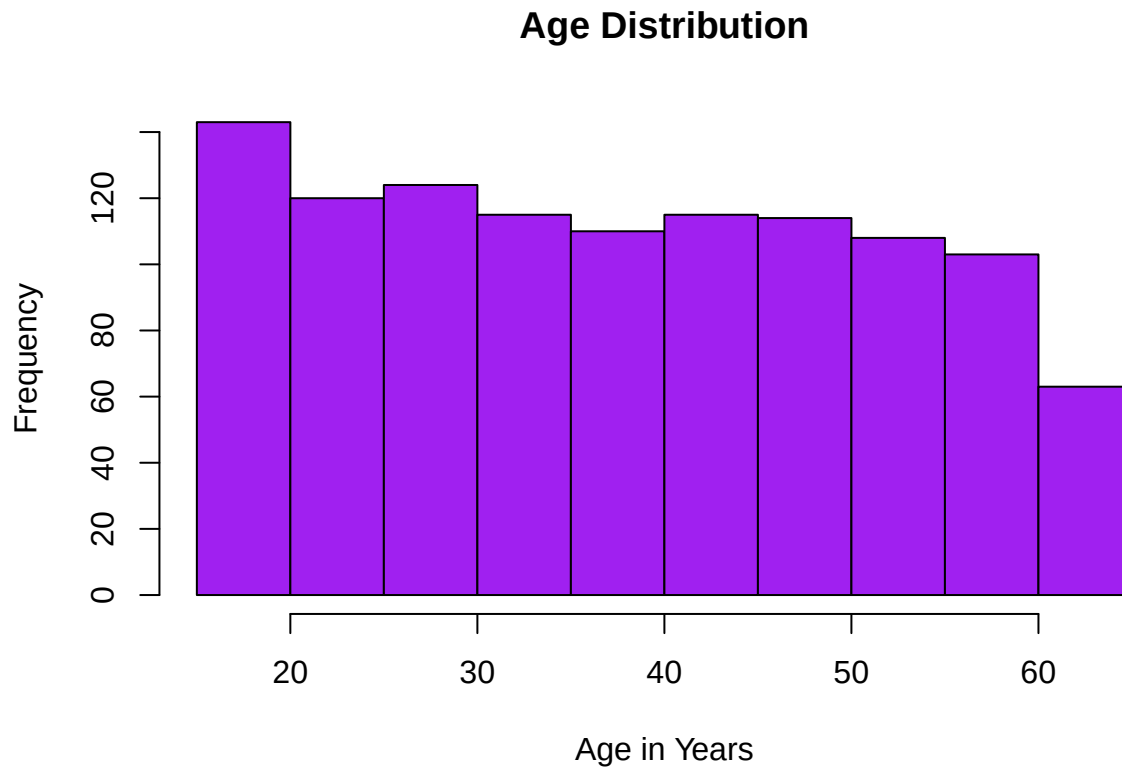
Summary Statistics

```
summary(r_cleaned)
```

```
##      age      sex      bmi      children
## Min.   :18.00 Length :1115 Min.   :15.96 Min.   :0.000
## 1st Qu.:26.00 N.unique : 2 1st Qu.:25.80 1st Qu.:0.000
## Median :38.00 N.blank  : 0 Median :29.81 Median :1.000
## Mean   :38.48 Min.nchar: 4 Mean   :30.16 Mean   :1.079
## 3rd Qu.:50.00 Max.nchar: 6 3rd Qu.:33.99 3rd Qu.:2.000
## Max.   :64.00      Max.   :53.13 Max.   :5.000
##      smoker      region      charges
## Length :1115 Length :1115 Min.   : 1122
## N.unique : 2 N.unique : 4 1st Qu.: 4150
## N.blank  : 0 N.blank  : 0 Median : 7730
## Min.nchar: 2 Min.nchar: 9 Mean   : 8590
## Max.nchar: 3 Max.nchar: 9 3rd Qu.:11880
##      Max.   :23401
```

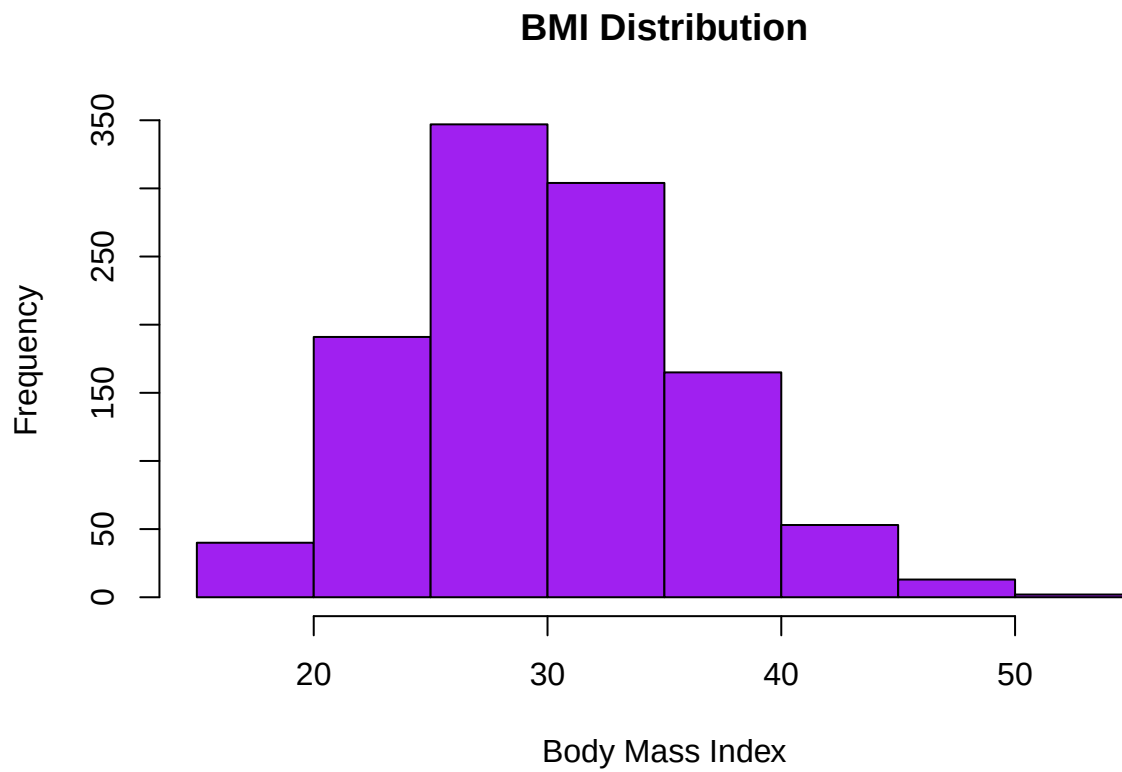
EDA

```
hist(r_cleaned$Age,  
     main = "Age Distribution",  
     xlab = "Age in Years",  
     col = "purple" )
```



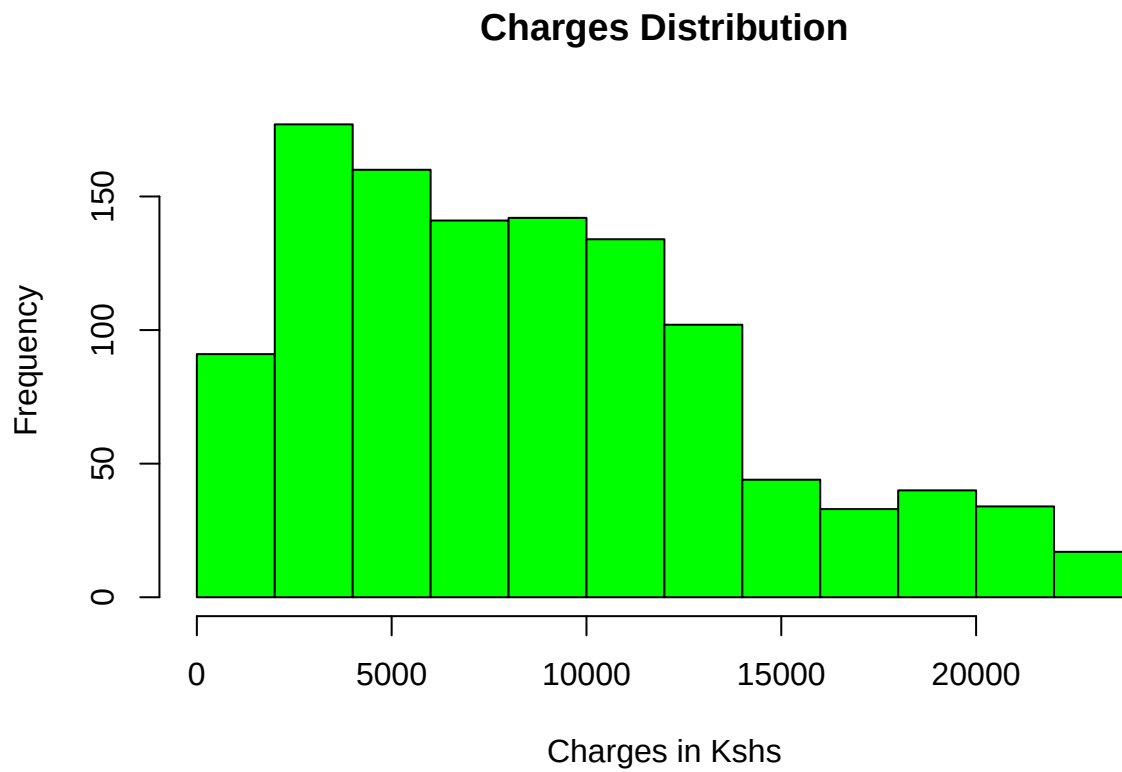
Age has a uniform distribution

```
hist(r_cleaned$bmi,  
     main = "BMI Distribution",  
     xlab = "Body Mass Index",  
     col = "purple" )
```



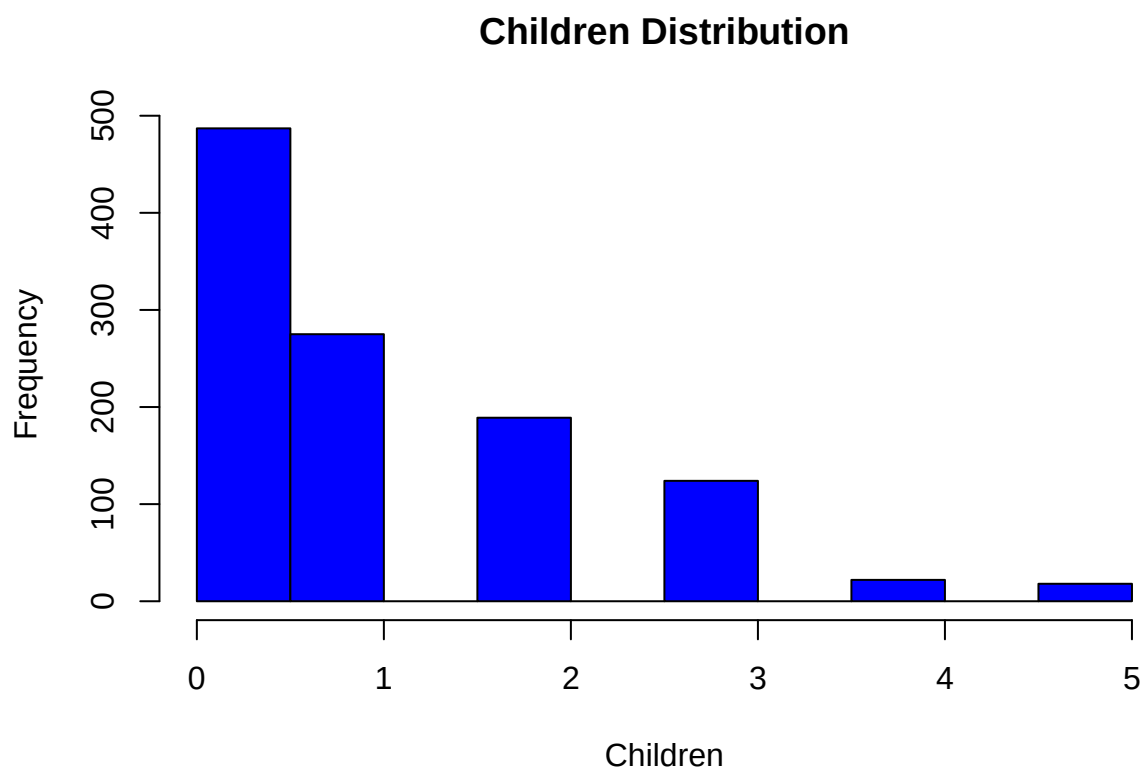
BMI distribution has a normal distribution

```
hist(r_cleaned$charges,  
     main = "Charges Distribution",  
     xlab = "Charges in Kshs",  
     col = "green" )
```



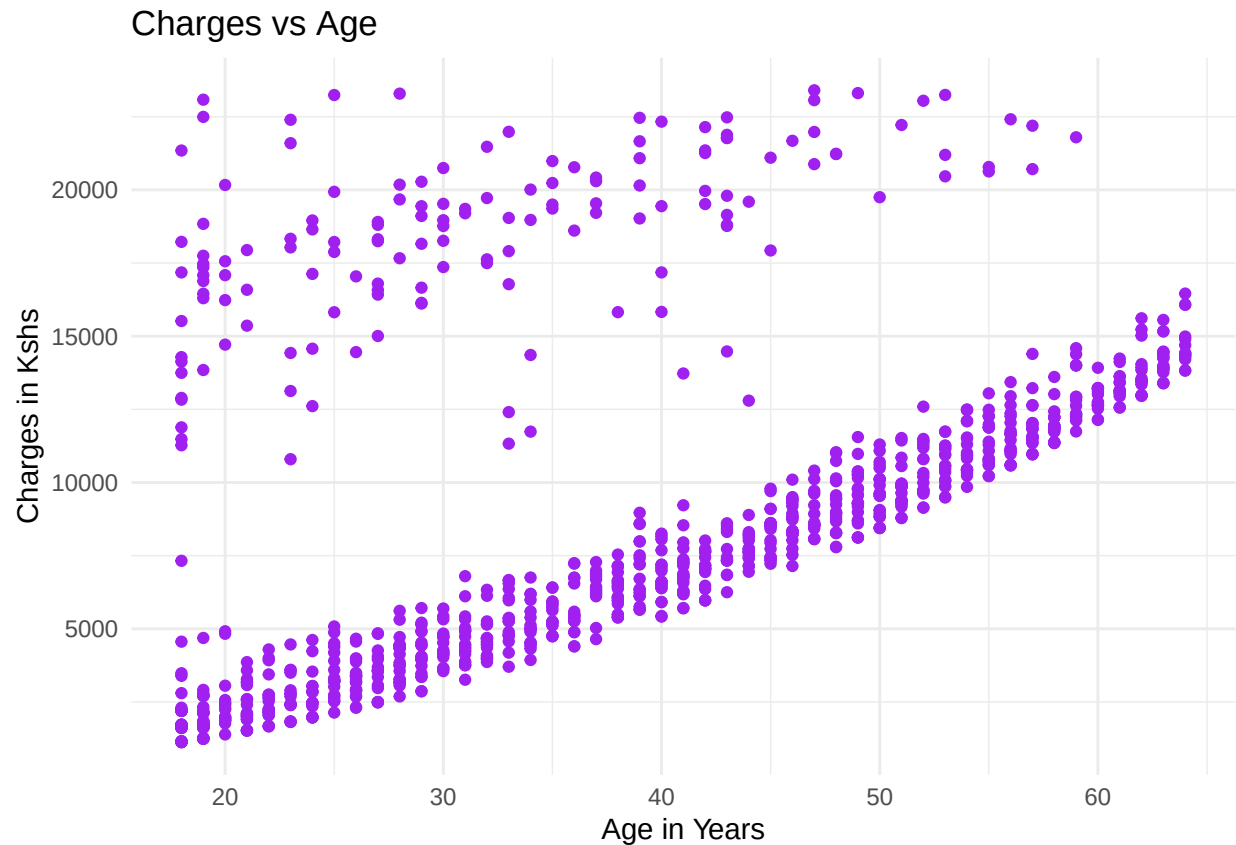
Charges distribution has a positively skewed distribution

```
hist(r_cleaned$children,  
     main = "Children Distribution",  
     xlab = "Children",  
     col = "blue" )
```



> The above histogram shows a positively skewed distribution

```
library(ggplot2)
ggplot(r_cleaned, aes(x = age, y = charges))+
  geom_point(color = "purple")+
  labs(title = "Charges vs Age", x = "Age in Years", y = "Charges in Kshs")+
  theme_minimal()
```



> The scatter plot indicates that age is a strong predictor of insurance charges implying as age increases charges also increases. The graph shows a clear positive relationship between age and charges, meaning costs tend to rise as people get older.

Data Manipulation

```
r_cleaned_filtered <- r_cleaned |>
dplyr::filter(age>25 , smoker=="no" , region=="northwest")
head(r_cleaned_filtered)
```

```
##   age    sex    bmi children smoker   region   charges
## 1  33  male  22.705         0     no northwest 21984.471
## 2  32  male  28.880         0     no northwest  3866.855
## 3  37 female  27.740         3     no northwest  7281.506
## 4  37  male  28.025         2     no northwest  6203.902
## 5  55 female  32.775         2     no northwest 12268.632
## 6  63  male  28.310         0     no northwest 13770.098
```

```
r_cleaned_selected <- r_cleaned |>
  dplyr::select(sex, charges, age )
head(r_cleaned_selected)
```

```
##      sex   charges age
```



```
## 1 female 16884.924 19
## 2 male 1725.552 18
## 3 male 4449.462 28
## 4 male 21984.471 33
## 5 male 3866.855 32
## 6 female 3756.622 31
```

```
r_cleaned_mutated <- r_cleaned|>
  dplyr :: mutate(charges_category=ifelse(charges >15000,"High","Low"))
head(r_cleaned_mutated)
```

```
##   age  sex   bmi children smoker   region   charges charges_category
## 1  19 female 27.900         0    yes southwest 16884.924           High
## 2  18  male 33.770         1     no southeast 1725.552            Low
## 3  28  male 33.000         3     no southeast 4449.462            Low
## 4  33  male 22.705         0     no northwest 21984.471           High
## 5  32  male 28.880         0     no northwest 3866.855            Low
## 6  31 female 25.740         0     no southeast 3756.622            Low
```

```
r_cleaned_arranged <- r_cleaned |>
  dplyr :: arrange(charges)
head(r_cleaned_arranged,20)
```

```
##   age sex   bmi children smoker   region   charges
## 1  18 male 23.21         0     no southeast 1121.874
## 2  18 male 30.14         0     no southeast 1131.507
## 3  18 male 33.33         0     no southeast 1135.941
## 4  18 male 33.66         0     no southeast 1136.399
## 5  18 male 34.10         0     no southeast 1137.011
## 6  18 male 34.43         0     no southeast 1137.470
## 7  18 male 37.29         0     no southeast 1141.445
## 8  18 male 41.14         0     no southeast 1146.797
## 9  18 male 43.01         0     no southeast 1149.396
## 10 18 male 53.13         0     no southeast 1163.463
## 11 19 male 19.80         0     no southwest 1241.565
## 12 19 male 20.30         0     no southwest 1242.260
## 13 19 male 20.70         0     no southwest 1242.816
## 14 19 male 27.60         0     no southwest 1252.407
## 15 19 male 28.70         0     no southwest 1253.936
## 16 19 male 30.40         0     no southwest 1256.299
## 17 19 male 34.10         0     no southwest 1261.442
## 18 19 male 34.40         0     no southwest 1261.859
## 19 19 male 35.40         0     no southwest 1263.249
## 20 20 male 33.33         0     no southeast 1391.529
```

```
r_cleaned |>
  dplyr :: group_by(sex) |>
  dplyr :: summarize(mean_charges=mean(charges,na.rm=TRUE))
```

```
## # A tibble: 2 x 2
##   sex    mean_charges
##   <chr>         <dbl>
```

```
## 1 female      8762.
## 2 male        8411.
```

```
r_cleaned |>
dplyr::group_by(smoker) |>
dplyr::summarize(mean_charges=mean(charges,na.rm=TRUE))
```

```
## # A tibble: 2 x 2
##   smoker mean_charges
##   <chr>      <dbl>
## 1 no         7660.
## 2 yes       18814.
```

```
r_cleaned |>
dplyr::group_by(region) |>
dplyr::summarize(mean_bmi=mean(bmi,na.rm=TRUE))
```

```
## # A tibble: 4 x 2
##   region    mean_bmi
##   <chr>      <dbl>
## 1 northeast    28.7
## 2 northwest    28.8
## 3 southeast    32.9
## 4 southwest    30.2
```

```
r_cleaned_processed <- r_cleaned|>
dplyr::filter(age > 25) |>
dplyr::select(region ,charges, age) |>
dplyr::mutate(age_group = ifelse(age > 25, "Adults", "Minor")) |>
dplyr::arrange(charges,10)
head(r_cleaned_processed)
```

```
##      region  charges age age_group
## 1 southwest 2302.300  26   Adults
## 2 southeast 2322.622  26   Adults
## 3 southeast 2483.736  27   Adults
## 4 southwest 2494.022  27   Adults
## 5 southeast 2497.038  27   Adults
## 6 southeast 2498.414  27   Adults
```

Data Visualization

Scatter Plot

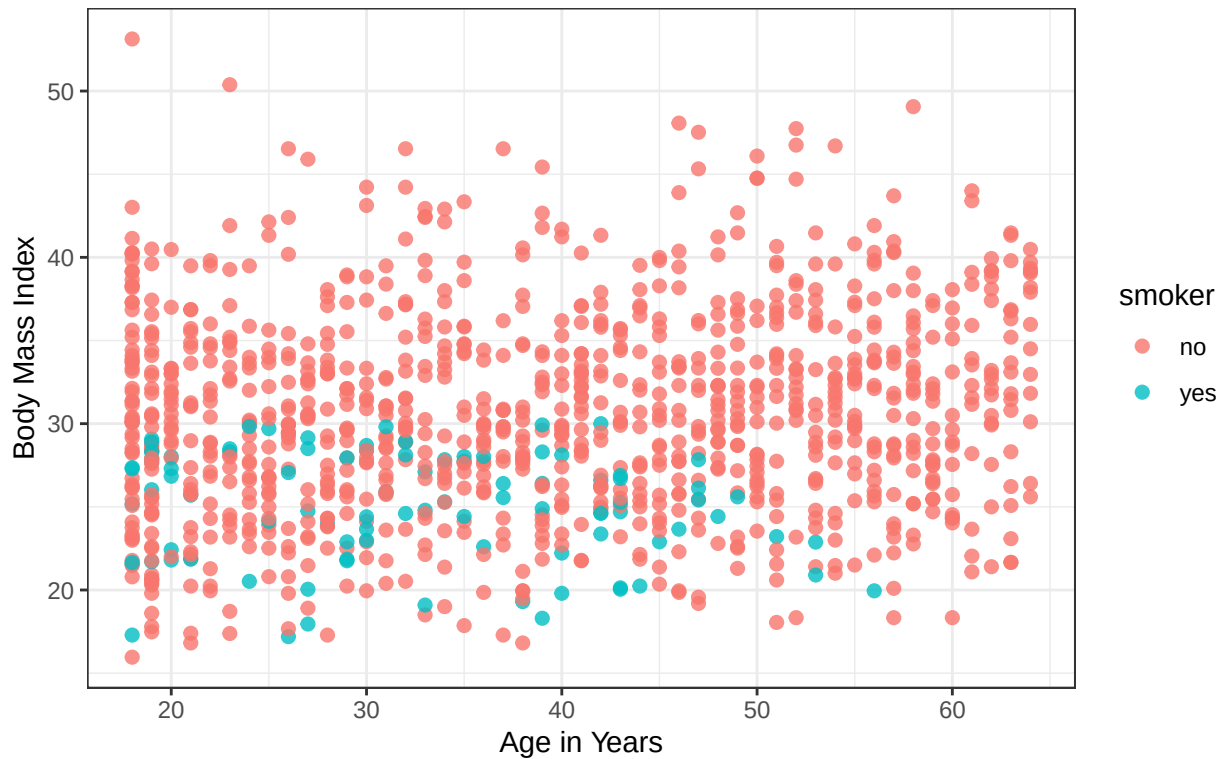
```
library(ggplot2)
r_cleaned |> ggplot (aes(x=age,y=bmi,color=smoker)) +
  geom_point(size = 2, alpha=0.8)+
  labs(x="Age in Years",
       y="Body Mass Index",
```

```

title="Scatter plot showing relationship between Age and BMI",
caption="Source:Insurance Dataset,2026")+
theme_bw()

```

Scatter plot showing relationship between Age and BMI



Source:Insurance Dataset,2026

BMI values are spread across all age groups, suggesting there is no strong relationship between age and BMI. Most individuals have BMI values between 20 and 40, and non-smokers appear to make up the majority of the data points. Smokers and non-smokers are distributed similarly across different ages and BMI ranges.

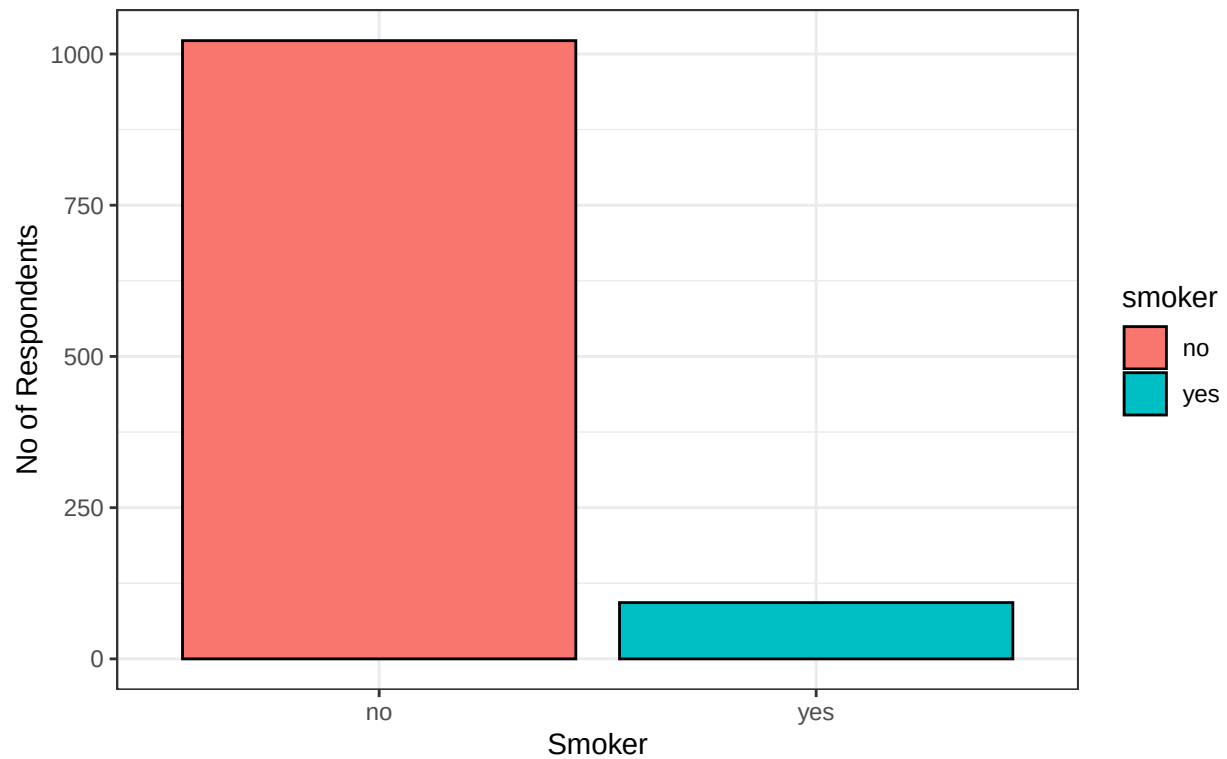
Bar Chart

```

r_cleaned |>
  ggplot(aes(x=smoker,fill=smoker))+
  geom_bar(color="black")+
  labs(x="Smoker",
       y="No of Respondents",
       title="A Bar Chart showing Smoker distribution",
       caption="Source: Insurance Dataset,2026")+
  theme_bw()

```

A Bar Chart showing Smoker distribution



Source: Insurance Dataset, 2026

Non-smokers make the largest number in the data set

Pie Chart

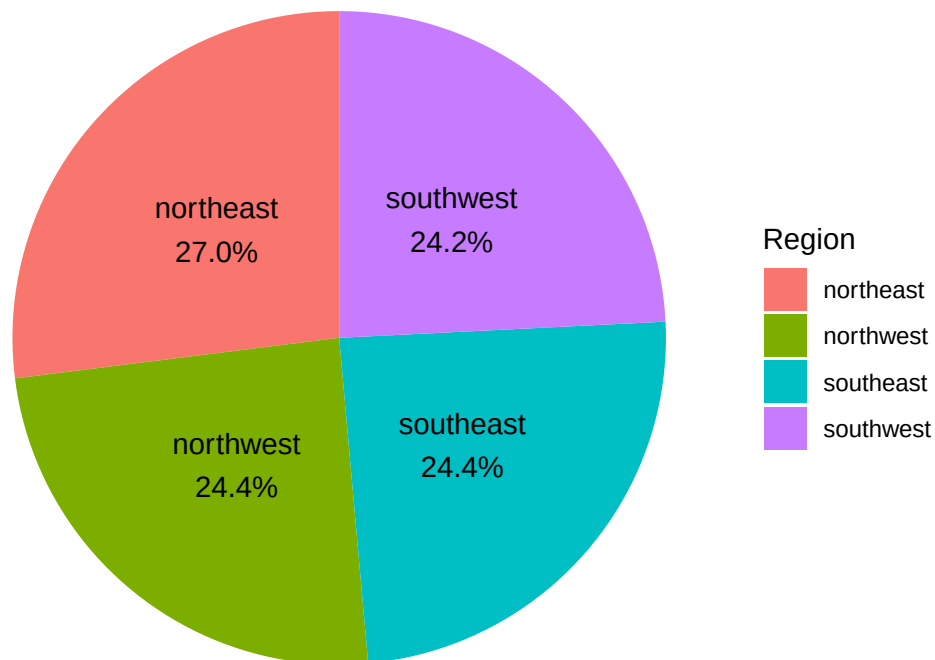
```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v lubridate  1.9.5      v tibble     3.3.1
## v purrr      1.2.2      v tidyr      1.3.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
r_cleaned |>
  group_by(region) |>
  summarise(total_charges = sum(charges, na.rm = TRUE)) |>
  mutate(
    pct = total_charges / sum(total_charges),
    label = paste0(region, "\n", scales::percent(pct, accuracy = 0.1))
  ) |>
  ggplot(aes(x = "", y = total_charges, fill = region)) +
```

```
geom_col(width = 1) +
geom_text(aes(label = label), position = position_stack(vjust = 0.5), size = 4) +
coord_polar(theta = "y") +
labs(
  title = "Total Charges by Region",
  fill = "Region"
) +
theme_void()
```

Total Charges by Region

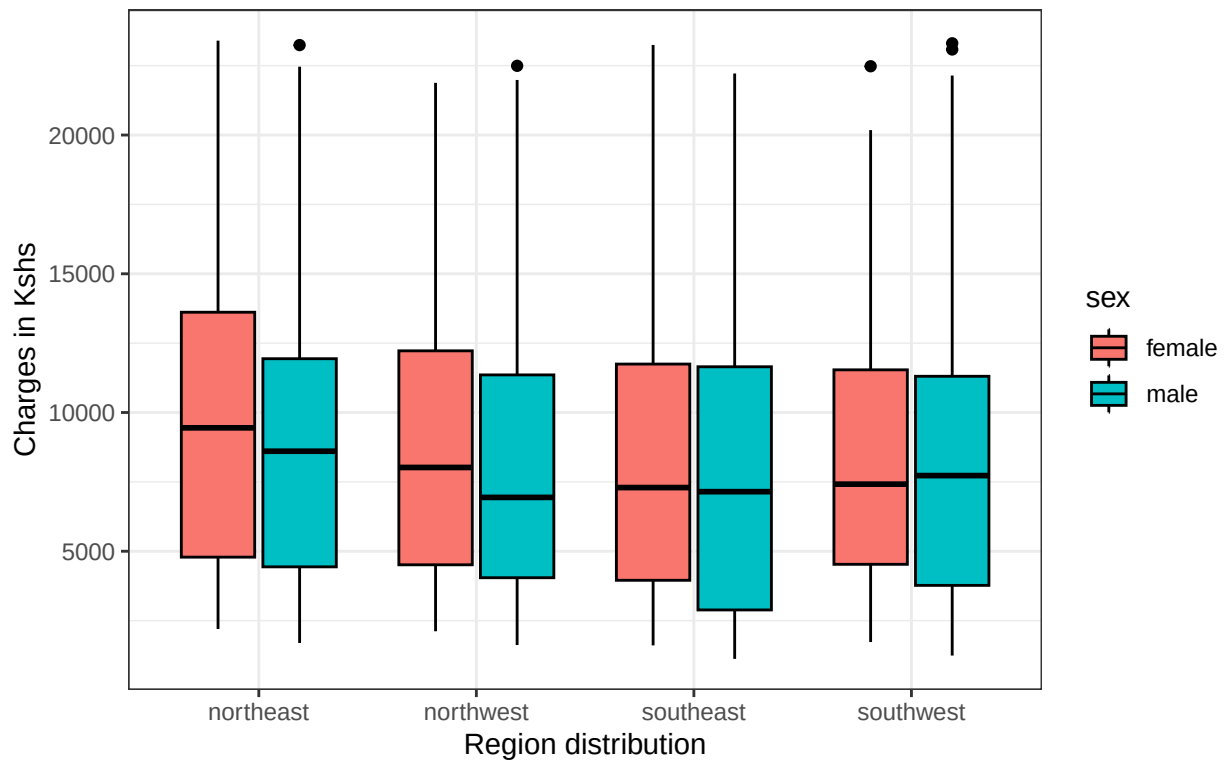


The northeast region accounts for the highest share of total charges at 27.0%. The northwest and southeast regions contribute equally at 24.4% each. The southwest region has the lowest share of total charges at 24.2%.

Boxplot

```
library(ggplot2)
r_cleaned |>
ggplot(aes(x=region,y= charges,fill=sex))+
geom_boxplot(color="black")+
labs(x="Region distribution",
  y="Charges in Kshs",
  title="A Boxplot of Charges by Region",
  caption="Source: Insurance Dataset,2026")+
theme_bw()
```

A Boxplot of Charges by Region



Source: Insurance Dataset,2026

The graph shows that insurance charges vary across regions and between males and females. The southeast region has the highest median charges for both sexes, while the northeast and northwest regions have relatively lower and similar distributions. Male charges tend to be slightly higher than female charges in most regions, and there are a few outliers indicating some individuals with high insurance Charges.

```
library(plotly)
```

```
##
## Attaching package: 'plotly'

## The following object is masked from 'package:ggplot2':
##
##   last_plot

## The following object is masked from 'package:stats':
##
##   filter

## The following object is masked from 'package:graphics':
##
##   layout
```

```
p=r_cleaned |>
  ggplot(aes(x= region, y = charges, fill = smoker)) +
  geom_boxplot() +
```

```

labs(x = "Region distribution",
     y = "Charges in Kshs",
     title = "Boxplot showing Charges Distribution by Region and Smoker",
     caption = "Source: Insurance Dataset, 2026") +
  theme_classic() +
  theme(legend.position = "bottom")

ggplotly(p)

```

Google Chrome was not found. Try setting the 'CHROMOTE_CHROME' environment variable to the executable

The boxplot shows that Insurance charges vary across all regions, but the distribution is quite similar. In every region, smokers tend to have much higher charges than non-smokers, as seen by the higher median values and wider spread in the upper box plots. There are also several outliers, indicating some individuals have extreme high medical expenses. Overall, smoking appears to have a stronger impact on charges than the region itself.

Hypothesis Testing

Two Sample t_test

HO: Mean insurance charges of Male is equal to mean insurance charges of Female
 H1: Mean insurance charges of Male is not equal to mean insurance charges of Female

```
result <- t.test(charges~sex, data=r_cleaned)
result
```

```
##
## Welch Two Sample t-test
##
## data: charges by sex
## t = 1.0779, df = 1102.5, p-value = 0.2813
## alternative hypothesis: true difference in means between group female and group male is not equal to 0
## 95 percent confidence interval:
## -287.4416 988.3175
## sample estimates:
## mean in group female mean in group male
## 8761.859 8411.421
```

From the results above where p-value is greater than 0.05, we fail to reject H_0 and conclude that the Mean Insurance charges of Male and Female is equal to each other

H_0 : There is no significant difference in mean Body Mass Index between Smokers and non-smokers H_1 : There is significant difference in mean Body Mass Index between Smokers and non-smokers

```
result <- t.test(bmi~smoker, data=r_cleaned)
result
```

```
##
## Welch Two Sample t-test
##
## data: bmi by smoker
## t = 14.438, df = 155.82, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group no and group yes is not equal to 0
## 95 percent confidence interval:
## 4.890029 6.440092
## sample estimates:
## mean in group no mean in group yes
## 30.63167 24.96661
```

The t-test results indicate a significant difference in mean Body Mass Index between smokers and non-smokers. The p-value is less than 0.05, indicating that the difference is statistically significant.

Chi Square Test

H_0 : There is no significant association between smoking status gender H_1 : There is significant association between smoking status gender

```
library(gtsummary)
tbl_summary(r_cleaned[c("sex", "smoker")],
            by="smoker") %>%
  add_ci() %>%
  add_p(
    test=list(
      all_categorical() ~ "chisq.test"
```


Characteristic	no N = 1,022 ¹	95% CI	yes N = 93 ¹	95% CI	p-value ²
sex					0.7
female	524 (51%)	48%, 54%	45 (48%)	38%, 59%	
male	498 (49%)	46%, 52%	48 (52%)	41%, 62%	

¹n (%)

²Pearson's Chi-squared test

Abbreviation: CI = Confidence Interval

```
)
)
```

Having my p-value>0.05 i fail to reject H0 at 5% Level of Significance and conclude that there is no significant association between smoking status and gender

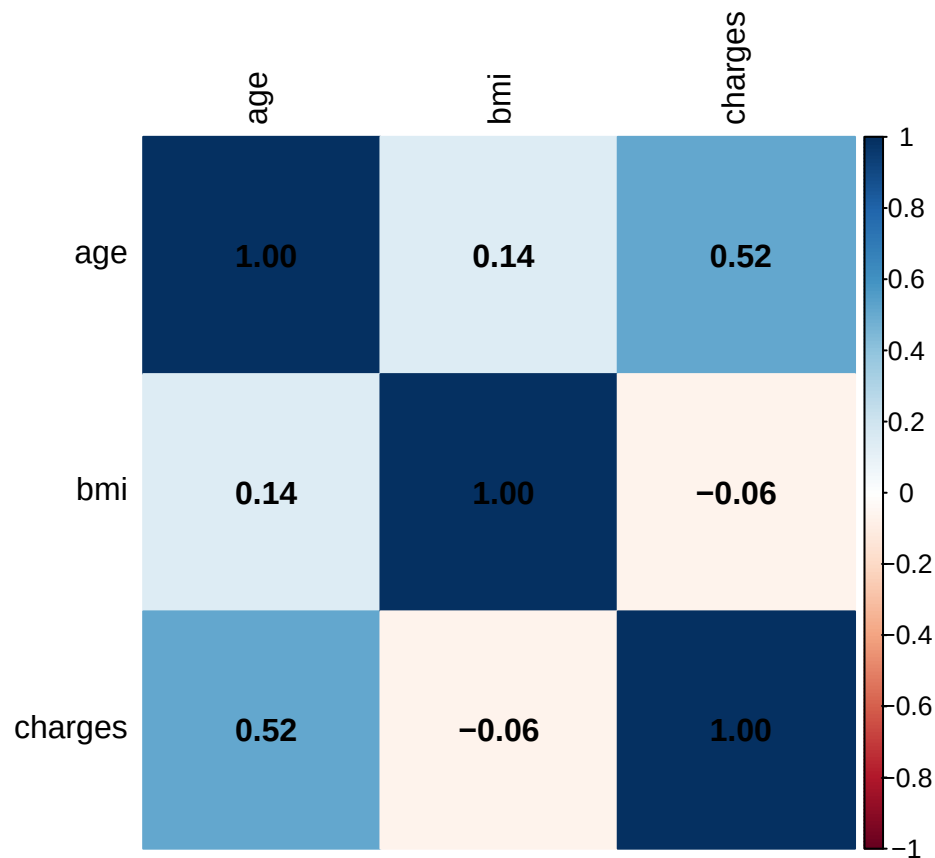
Correlation Analysis

Heatmap

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
r_cleaned |>
  select(age,bmi,charges) |>
  cor(use="complete.obs") |>
  corrplot(method = "color",
           addCoef.col = "black",tl.col = "black")
```



The heatmap shows that age has a moderate positive relationship with charges , meaning insurance costs tend to rise as a person gets older. In contrast, there is a very weak positive relationship between age and BMI , and a negative relationship between BMI and charges ,indicating that BMI alone does not strongly correlate with changes in charges or age in this dataset.

Regression Analysis

Simple Linear Regression

H0:Age has no significant effect on charges H1: Age has a significant effect on charges

```
model <- lm(charges ~ age, data = r_cleaned)
summary(model)

##
## Call:
## lm(formula = charges ~ age, data = r_cleaned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3809.0 -2630.4 -1654.2  -194.1 18452.7
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 768.49 409.48 1.877 0.0608 .
## age 203.25 10.01 20.302 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4633 on 1113 degrees of freedom
## Multiple R-squared: 0.2703, Adjusted R-squared: 0.2696
## F-statistic: 412.2 on 1 and 1113 DF, p-value: < 2.2e-16
```

From the results my p-value is less than 0.05 at 5% level of significance, therefore conclude that age has a significant effect on charges. As one unit of age increases charges by 203.25 units.

Multiple Linear Regression

H0: Age, BMI, Children, Sex, Smoker do not have a significant effect on charges
H1: At least one of the variables has a significant effect on charges

```
model <- lm(charges ~ age+bmi+children+sex+smoker, data = r_cleaned)
summary(model)
```

```
##
## Call:
## lm(formula = charges ~ age + bmi + children + sex + smoker, data = r_cleaned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3622.5 -1199.9  -774.9   -35.0  20704.3
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2274.967    528.660  -4.303 1.83e-05 ***
## age          234.815     6.592   35.620 < 2e-16 ***
## bmi           15.965    15.358    1.040  0.2988
## children     409.989    73.903    5.548 3.62e-08 ***
## sexmale     -336.307   179.949   -1.869  0.0619 .
## smokeryes   12823.326   338.121   37.925 < 2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3001 on 1109 degrees of freedom
## Multiple R-squared: 0.6949, Adjusted R-squared: 0.6935
## F-statistic: 505.1 on 5 and 1109 DF, p-value: < 2.2e-16
```

The model expresses charges by 69.49 percentage. From the results p-value of Age is less than 0.05 at 5% level of significance, therefore conclude that age has a significant effect on charges. As one unit of age increases charges by 203.25 units. As Body Mass Index increases by a unit it increases charges by 15.965 units. Since p-value is greater than 0.05, it indicates that body mass index do not have a significant effect on charges. For each additional child, charges increases by 409.989 units. Since p-value is less than 0.05, it indicates that children have a significant effect on charges. Male have 336.307 unit charges lower than females. Since p-value is greater than 0.05, it indicates that male do not have a significant effect on charges. Smokers pay charges by 12823.326 units more than non-smokers. Since p-value is less than 0.05, it indicates that male have a significant effect on charges. Thus, reject H0 and conclude that at least one of the variables has a significant effect on charges

**SPECIAL THANKS TO CENTER FOR DATA ANALYTICS
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**CERTIFICATION COURSE:INTRODUCTION TO R FOR DATA
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